UNIVERSITY *Of* KIGALI *"Unequalled Education Excellence"*

FINAL EXAMINATION

GRADUATE SCHOOL

PROGRAMME:
MSC-INFORMATION TECHNOLOGY

MODULE CODE AND TITLE: MIT91205: BIG DATA ANALYTICS	**LEVEL AND TRIMESTER:** LEVEL 9, TRIMESTER 3
DATE: 26/07/2026	**TIME:** 9:30 AM - 12:30 AM **SESSION:** WEEKEND
DURATION: 3 HOURS	**TOTAL MARKS:** 100 TO BE SCALED TO 40 MARKS
EXAMINER/ LECTURER: DR. MUSABE Jean Bosco & DR. BUGINGO EMMANUEL	**MODERATOR:** Dr Wilson Musoni

INSTRUCTIONS TO THE CANDIDATE

1. Questions must be attempted in the answer booklet provided.
2. All queries should be directed to the invigilator; do not communicate or attempt to communicate with any other candidate.
3. You have **THREE HOURS** to complete this paper. You are not allowed to leave the examination room within the first hour and in the last 15 minutes of this examination.
4. This is a **CLOSED BOOK** examination.
5. Answer 4 Questions; Question 1 (Section A) is compulsory, and choose any 3 other questions in Section B.
6. Please do not write on the question paper.

SECTION A (COMPULSORY QUESTION) 40 MARKS

QUESTION ONE

Social Trend is a social media analytics company that provides real-time sentiment analysis and trend detection services to brands, political campaigns, and market research firms. The company processes data from multiple social media platforms, including Twitter, Facebook, Instagram, Reddit, and YouTube, ingesting approximately 500 million posts per day, which represents about 5,800 posts per second on average, with peak volumes exceeding 20,000 posts per second during major events. The company's current data processing system is based on batch processing, which means that sentiment analysis results are available with a delay of several hours. This delay is unacceptable for many of their clients, including political campaigns that need to respond to public sentiment in real-time, brands that need to manage reputation crises immediately, and financial firms that use sentiment analysis for algorithmic trading decisions. The Chief Executive Officer, Ms. Grace Mwangi, has decided to implement a real-time sentiment analysis pipeline that can provide results with a latency of less than five seconds. The pipeline must ingest streaming data from multiple social media APIs, preprocess and clean the text data, apply sentiment analysis models, detect trending topics and emerging trends, and deliver results to clients through APIs, dashboards, and alert systems. The system must handle the peak data volumes while maintaining low latency, scale horizontally as the company grows, and provide exactly-once processing guarantees to ensure data accuracy.

Required:

a. Explain the architecture of Apache Kafka, including brokers, topics, partitions, producers, and consumers, and describe how Social Trend would structure their topics for different social media platforms and data types. Your response should demonstrate a clear understanding of how to design a multi-platform Kafka architecture that handles different data formats, rates, and priorities.

(10 Marks)

b. Call the concept of consumer groups in Kafka and explain how this would enable Social Trend to process twenty thousand posts per second during peak events with sub-five-second latency. Your response should address the relationship between partitions and consumers across different platforms, how parallelism is achieved, and how consumer group rebalancing would work when processing volumes fluctuate.

(10 Marks)

c. Elaborate on the three processing guarantees available in Kafka and analyze which guarantee Social Trend should implement for their sentiment analysis pipeline, providing detailed justification based on the business context of social media analytics where accuracy and timeliness are both critical. Your response should consider the consequences of each guarantee in terms of analysis accuracy and client satisfaction.

(10 Marks)

d. Discuss the distinction between Event Time and Processing Time in Spark Structured Streaming and explain why Event Time is essential for Social Trend's sentiment analysis where social media posts may experience delays due to API rate limiting, network issues, or platform-specific delays. Your response should explain how Event Time enables accurate trend detection and sentiment analysis that reflects actual public sentiment rather than processing delays. **(10 Marks)**

20MARKS

QUESTION TWO

HealthData Systems is a healthcare technology company that provides electronic health record platforms and healthcare analytics solutions to hospitals, clinics, and healthcare networks across fifty countries. The company's platform manages data for approximately 100 million patients worldwide, including comprehensive medical histories, diagnostic images, laboratory results, genomic sequences, clinical trial data, and real-time patient monitoring from wearable devices. The company's current platform relies heavily on a monolithic relational database that is struggling to meet the growing demands of modern healthcare data management. Query performance has degraded significantly as the database has grown, with some complex clinical queries taking over thirty seconds to return results. The rigid relational schema cannot accommodate the evolving and diverse data structures required for genomic data and real-time monitoring. Scalability limitations are preventing the company from expanding into new markets, and infrastructure costs are growing at an unsustainable rate. The Chief Technology Officer, Dr. Mark Ochieng, has concluded that the company must adopt a polyglot persistence strategy using multiple NoSQL databases to address these challenges. The strategy must support the diverse data types present in healthcare while maintaining compliance with stringent healthcare data regulations such as HIPAA in the United States, GDPR in Europe, and PDPA in Asia. The solution must also ensure data security, patient privacy, and auditability.

Required:

a. Clarify the concept of polyglot persistence and analyze why HealthData Systems would benefit from using multiple database types rather than maintaining their single relational database approach. Your response should demonstrate an understanding of the strengths and weaknesses of different database technologies and explain how a polyglot approach addresses the specific challenges of healthcare data management. **(10 Marks)**

b. Compare the four types of NoSQL databases, specifically Key-Value stores, Document databases, Column-Family databases, and Graph databases, providing HealthData Systems use cases and specific technology examples for each type. Your response should address how each database type would serve different healthcare data needs, such as patient profiles, genomic data, clinical trial tracking, and medical imaging metadata. **(10 Marks)**

QUESTION THREE

InsuranceGuard is a global insurance company that processes millions of claims annually across healt auto, property, and life insurance lines. The company is facing significant challenges with fraud detection, with approximately ten percent of claims containing fraudulent elements, costing the company approximately one billion dollars annually. Current fraud detection systems use rules-based approaches and statistical models that identify individual suspicious claims but fail to detect organized fraud rings and complex fraud networks. The Chief Fraud Investigator, Mr. James Omondi, has concluded that the company needs to adopt graph-based analytics to detect fraudulent networks and patterns that are invisible to traditional approaches. Fraud rings often involve multiple individuals working together, with patterns such as shared addresses, shared phone numbers, shared medical providers, or shared repair shops. These connections can be modeled as a graph where individuals, providers, and locations are nodes, and relationships such as shared addresses or transactions are edges. The Head of Data Science, Dr. Grace Akinyi, has proposed using Spark GraphX to implement graph-based fraud detection. The solution must identify fraud rings and suspicious patterns in real-time, scale to handle the company's growing data volumes, integrate with existing fraud detection systems, and provide explainable results that can be used in legal proceedings against fraudsters.

Required:

a. Elucidate the core concepts of Spark GraphX, including VertexRDD, EdgeRDD, and Graph objects, and describe how InsuranceGuard would model their claims data as a graph for fraud detection. Your response should demonstrate a clear understanding of graph data structures and explain how nodes and edges would represent different entities and relationships in the insurance claims context.

(10 Marks)

b. Describe the PageRank algorithm in GraphX and explain how InsuranceGuard would use PageRank to identify the most influential entities in their fraud networks. Your response should explain how PageRank works, what it measures, and how high PageRank scores might indicate important entities in fraud networks such as central organizers or frequently used service providers.

(10 Marks)

QUESTION FOUR

20MARKS

GlobalHealth Analytics is a multinational healthcare data analytics company that provides insights and predictive models to healthcare providers, insurance companies, pharmaceutical companies, and government health agencies across sixty countries. The company processes data from over 2,000 hospitals, 10,000 clinics, and 50 million patients, representing one of the largest healthcare datasets globally. The company handles a diverse range of healthcare data including electronic health records

with comprehensive medical histories, clinical notes containing unstructured text, medical imaging data such as X-rays and MRIs, genomic sequencing data, real-time patient monitoring from wearable devices, and clinical trial results and drug efficacy data. This data is subject to stringent regulatory requirements including the General Data Protection Regulation in Europe, the Health Insurance Portability and Accountability Act in the United States, the Personal Data Protection Act in various Asian jurisdictions, and multiple local and national healthcare data regulations. GlobalHealth is facing serious challenges that threaten their business operations and reputation. The company experienced two significant data breaches in the past eighteen months, resulting in substantial financial losses and reputational damage. Compliance violations have resulted in fines totaling over seventy-five million dollars across multiple jurisdictions. Annual cloud infrastructure costs have exceeded thirty-five million dollars and continue to grow at approximately twenty percent annually. Data quality issues affect approximately twelve percent of clinical records, undermining the accuracy of their analytics. Audit response times average four months, creating substantial regulatory risk and delaying essential reporting to health authorities. The Chief Information Officer, Dr. Grace Akinyi, has asked you to design a comprehensive data governance and security framework that addresses all these challenges while supporting the company's continued growth and innovation in healthcare analytics.

Required:

a. Explain the key pillars of data governance and describe how GlobalHealth would implement each pillar in their healthcare data platform to ensure comprehensive data management, regulatory compliance, and data trustworthiness. Your response should address the specific challenges of healthcare data, including the need for rigorous patient privacy protections, the importance of accurate clinical data for patient safety, the requirement for comprehensive audit trails, and the need to balance data accessibility with security. **(8 Marks)**

b. Describe the six dimensions of data quality, including accuracy, completeness, consistency, timeliness, validity, and uniqueness, and explain how GlobalHealth would ensure each dimension for their diverse healthcare data sources. Your response should provide practical approaches to data quality management in the context of large-scale healthcare analytics, including specific techniques for validating clinical data, detecting data anomalies, and maintaining data integrity across multiple systems. **(12 Marks)**

QUESTION FIVE

CloudTech Solutions is a cloud consulting firm that helps organizations optimize their cloud infrastructure costs. The company has observed that many organizations are overspending significantly on their cloud big data platforms, with costs often exceeding budgets by fifty percent or more. These

cost overruns are driven by several factors including inefficient storage utilization, over-provisi compute resources, lack of cost visibility, and suboptimal query patterns. The company has be engaged by several clients to help them optimize their cloud big data costs. These clients use various cloud platforms including AWS, Google Cloud, and Azure, and they run a variety of big data workloads including data lakes, data warehouses, real-time streaming, and machine learning. Each client has unique requirements but shares the common goal of reducing costs while maintaining or improving performance. The Chief Cloud Architect, Ms. Grace Ochieng, has asked you to develop a comprehensive cost optimization framework for cloud big data platforms that addresses the various cost drivers and provides practical strategies for reducing costs. The framework must be applicable across different cloud platforms and workload types, and it must balance cost reduction with performance and reliability requirements.

Required:

a. Clarify the key cost drivers in cloud big data platforms, including storage costs, compute costs, and data transfer costs, and analyze how each driver contributes to overall cloud spending. Your response should address the specific characteristics of big data workloads that make them expensive to run in the cloud.

(10 Marks)

b. Describe storage optimization strategies for cloud big data platforms, including storage tiering, data compression, data lifecycle management, and data deduplication, and explain how these strategies reduce storage costs. Your response should address how different data types and access patterns should be managed to optimize storage costs.

(10 Marks)

QUESTION SIX

20MARKS

FinSecure is a multinational financial services corporation that processes millions of banking and investment transactions daily across retail banking, investment banking, and wealth management divisions. The company is currently facing significant challenges in fraud detection, customer churn prevention, and risk assessment due to the limitations of their traditional analytics platforms.

Fraud detection currently uses rules-based systems that are easily circumvented by sophisticated criminals, with approximately fifteen percent of fraudulent transactions going undetected. Customer churn prediction models rely on outdated methods that identify at-risk customers only after they have already initiated the process of leaving the bank. Risk assessment models take over 48 hours to generate credit scores, limiting the bank's ability to approve loans quickly in competitive markets. These challenges are costing the company substantial revenue and putting them at a competitive disadvantage.

The Chief Data Scientist, Dr. Sarah Kilonzo, has decided to implement a comprehensive machine learning platform using Spark MLlib for scalable model training and deployment. The platform must provide sub-second scoring latency for transaction fraud detection, support daily model retraining to adapt to evolving fraud patterns, enable real-time customer churn prediction, and scale to accommodate the company's growing data volumes.

Required:

a. Describe the feature engineering pipeline in Spark MLlib, including the roles of Vector Assembler, StandardScaler, and StringIndexer, and explain how these components would be applied to FinSecure's transaction data to prepare it for machine learning modeling. Your response should address the specific challenges of financial data, including handling categorical variables such as merchant categories and transaction types, normalizing numerical features such as transaction amounts and account balances, and dealing with missing or incomplete data. **(10 Marks)**

b. Discuss the importance of feature engineering in machine learning and explain how FinSecure would use Spark MLlib's feature transformers to handle categorical variables, normalize numerical features, and combine features for their fraud detection models. Your response should address the specific challenges of financial data and explain how proper feature engineering improves model performance. **(10 Marks)**

END

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FINAL EXAMINATION

GRADUATE SCHOOL

PROGRAMME: MSC-INFORMATION TECHNOLOGY
SESSION: WEEKEND

MODULE CODE AND TITLE: MSCIT 9128: MACHINE LEARNING AND BIG DATA ANALYTICS	**LEVEL AND TRIMESTER:** LEVEL 9, TRIMESTER 2
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EXAMINER/ LECTURER: Dr. BUGINGO EMMANUEL	**MODERATOR:** DR. BOSCO MUSABE

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SECTION A (COMPULSORY QUESTION) 40 MARKS

QUESTION ONE

In an effort to promote sustainable food production, a regional agricultural board has launched a Big Data–driven precision farming initiative. The project seeks to optimize resource use and enhance crop yield prediction across large-scale farms using integrated IoT, remote sensing, and analytics technologies. Sensors deployed throughout the fields continuously collect data on soil nutrients, temperature, moisture levels, and pest activity, while drones and satellites provide high-resolution imagery of crop health.

All incoming data is stored and processed through distributed systems such as Hadoop Distributed File System (HDFS) and cloud-based platforms like Amazon S3 to handle the massive volume, velocity, and variety of agricultural data. To enable near real-time insights, Apache Spark supports in-memory analytics, while Apache Kafka manages live data streams from IoT devices.

The analytical phase involves predictive models to forecast yield and disease outbreaks, while prescriptive analytics guides optimal irrigation schedules and fertilizer application. By combining machine learning with spatial and temporal data, the project aims to create an intelligent decision support system that assists farmers and policymakers alike. Nevertheless, the team must address data quality challenges arising from faulty sensors, inconsistent satellite feeds, and varying environmental conditions. Ensuring veracity and reliability of agricultural data is crucial for accurate decision-making and long-term sustainability.

Required:

1. Identify the 5Vs of Big Data present in this project. **(5 Marks)**
2. Describe how data sources, storage, processing, and analysis can be integrated into the city's Big Data architecture. **(6 Marks)**
3. Compare the suitability of Apache Hadoop and Apache Spark for this project. **(6 Marks)**
4. Illustrate how Kafka could be implemented to manage real-time traffic updates. **(5 Marks)**
5. Analyze the role of predictive and prescriptive analytics in improving city operations. **(6 Marks)**
6. Evaluate the potential risks of data veracity and recommend methods to enhance data quality. **(6 Marks)**
7. Propose one innovative application of spatial analytics in this smart city project. **(6 Marks)**

x dynamic curb space

x time stamp
x drive malfunction
x

SECTION B (SELECT ONLY THREE) 60 MARKS

QUESTION TWO

A local bank wants to detect unusual transactions that could indicate potential fraud. The bank processes thousands of transactions daily, but fraudulent ones are rare. Machine learning is being used to analyze transaction patterns and raise alerts for suspicious activities.

Required:

1. Explain why stream processing might be preferred over batch processing. **(4 marks)**
2. Classify this task as classification or regression and justify. **(4 marks)**
3. Identify two evaluation metrics suitable for imbalanced data. **(4 marks)**
4. Discuss how real-time alerts could reduce losses. **(4 marks)**
5. Suggest one strategy to reduce false alarms in fraud detection. **(4 marks)**

QUESTION THREE

An online learning platform collects student engagement data such as login frequency, video completion rate, and quiz performance. The institution wants to use analytics to predict dropout risk and recommend learning support.

Required:

1. Describe how predictive analytics could identify at-risk students. **(4 marks)**
2. Compare descriptive and predictive analytics in this context. **(4 marks)**
3. Suggest one data visualization tool for tracking learning progress. **(4 marks)**
4. Explain how ethical considerations apply when analyzing student data. **(4 marks)**
5. Propose ways to balance model accuracy and fairness. **(4 marks)**

QUESTION FOUR

A hospital uses machine learning models to analyze X-ray images for early disease detection. They plan to combine structured data (age, gender, symptoms) with image data to enhance diagnostic accuracy.

Required:

1. Explain how deep learning can assist in image recognition tasks. **(4 marks)**
2. Describe how data preprocessing improves image quality before modeling. **(4 marks)**

reliable

3. Identify challenges of model interpretability in healthcare. **(4 marks)**

4. Discuss how bias might appear in medical datasets. **(4 marks)**

5. Suggest techniques to ensure fairness and transparency. **(4 marks)**

QUESTION FIVE

A retail company wants to optimize its inventory and predict demand across multiple branches. It plans to use sales data, weather conditions, and promotional events for analysis.

Required:

1. Explain how temporal analytics could improve demand forecasting. **(4 marks)**

2. Describe how spatial analytics can help identify regional demand patterns. **(4 marks)**

3. Differentiate between operational and descriptive analytics in this case. **(4 marks)**

4. Propose how real-time dashboards could support decision-making. **(4 marks)**

5. Evaluate the limitations of relying solely on analytics for strategic decisions. **(4 marks)**

QUESTION SIX

A local bank is on a mission to enhance its security measures by detecting unusual transactions that might signal potential fraud. With thousands of transactions processed each day, the bank faces the challenge of identifying rare fraudulent activities amidst an ocean of legitimate ones. To tackle this issue, the bank is employing machine learning techniques to meticulously analyze transaction patterns. This advanced technology enables the bank to raise alerts for any activities that appear suspicious, allowing for quicker response and prevention of potential financial crime.

Required:

1. Explain how sentiment analysis extracts opinions from social media data. **(4 marks)**

2. Discuss the benefits of combining text analytics and visualization tools. **(4 marks)**

3. Suggest how real-time analytics can enhance marketing responses. **(4 marks)**

4. Assess challenges posed by fake accounts and misinformation. **(4 marks)**

5. Recommend measures to improve data reliability. **(4 marks)**

END

Final exam past papers (questions and answers)

MIT91205 Final Exam, July 2026: Model Answers

Format: 3 hours, 100 marks. Question 1 compulsory (40 marks); choose 3 from Section B (20 marks each).

Question 1 (40 marks): Real-time sentiment pipeline with Kafka and Spark

Key facts: 500 million posts a day (about 5,800 per second, over 20,000 per second at peak), 5 platforms, results in under 5 seconds, exactly-once.

(a) Kafka architecture and topic design (10 marks)

- **Broker**: a Kafka server that stores partitions. Start with 5 to 6 brokers.
- **Topic**: a named stream of records. **Partition**: an ordered log; the unit of parallelism.
- **Producer** writes records (partition chosen by hashing the key); **consumer** reads and tracks its offset.
- **Replication**: factor 3 with `min.insync.replicas=2`, so one broker can fail with no data loss.
- **One raw topic per platform**: each API has its own format and rate, and an outage on one does not slow the others.
- **Shared cleaned topic** with one Avro schema, checked by a schema registry.
- **Partitions**: 20,000 posts/s at about 1,000 per consumer needs at least 20; plan 24 to 48 for busy topics.
- **Keys and retention**: key raw posts by `post_id`, scores by brand; keep raw data 7 days for replay.

```
social.twitter.raw  social.facebook.raw  ... (one per platform)
        |  Spark cleaning
social.all.cleaned  (Avro schema)
        |  Spark sentiment
sentiment.scores ---> dashboards, APIs
        |  Spark trends
alerts.trending  ---> alert service
```

Remember: partitions, replication 3, topic per platform, schema registry.

(b) Consumer groups and parallelism (10 marks)

- **Consumer group**: consumers that share a topic; each partition goes to exactly one consumer in the group.
- **Max parallelism = partition count**: 24 partitions allow 24 consumers; a 25th sits idle.
- **Sizing**: 24 consumers handle about 830 posts/s each at peak, keeping latency well under 5 seconds.
- **Several groups** (sentiment, trends, archive) read the same topic independently, each with its own offsets.
- **Rebalancing**: when a consumer joins, leaves or crashes, its partitions move to the others.
- **No skipped data**: the new owner resumes from the last committed offset.
- **Cooperative sticky assignor**: only the moved partitions pause, so the group keeps working.
- **Consumer lag** (latest offset minus committed offset) rising is the signal to add consumers.

Remember: one partition per consumer, lag, rebalancing, cooperative sticky.

(c) Processing guarantees (10 marks)

Guarantee	Offset committed	On failure	Effect
At-most-once	before processing	message lost	sentiment under-counts
At-least-once	after processing	message repeated	duplicates, false alerts
Exactly-once	with the output, atomically	no loss, no duplicates	correct counts

- **Idempotent producer** (`enable.idempotence=true`) drops duplicate retries.
- **Transactions** write output records and consumer offsets together, or not at all.
- `read_committed` consumers never see aborted records.
- **Spark**: checkpointing, a replayable source (Kafka) and an idempotent sink give end-to-end exactly-once.
- **Choose exactly-once**: clients trade and react to alerts, so duplicates or losses cost money and trust.
- **Cost**: a few tens of milliseconds of latency, which still fits the 5 second target.

Remember: idempotence, transactions, read_committed, checkpoint.

(d) Event time vs processing time (10 marks)

- **Event time**: when the post was written. **Processing time**: when Spark handles it.
- **Late data**: API rate limits, network problems and retries delay posts by seconds or minutes.
- **Correct windows**: event time counts a post in the minute it was written.
- **No fake spikes**: a throttled backlog arriving at once is spread back over the right minutes.
- **Reproducible**: replaying from Kafka gives the same results.
- **Watermark**: how long to wait for late posts; longer is more accurate but slower and uses more memory.

```
trend = (parsed
    .withWatermark("event_time", "2 minutes")
    .groupBy(F.window("event_time", "1 minute", "30 seconds"), "topic")
    .agg(F.count("*").alias("mentions"), F.avg("sentiment").alias("avg_sentiment")))
query = (trend.writeStream.outputMode("update")
    .option("checkpointLocation", "/chk/trends")
    .trigger(processingTime="2 seconds").format("console").start())
```

Remember: event time, watermark, window, late data.

Question 2 (20 marks): Polyglot persistence for healthcare

Key facts: 100 million patients, 50 countries, one relational database, queries over 30 seconds, HIPAA, GDPR and PDPA.

(a) Polyglot persistence (10 marks)

- **Definition**: use several database types, each for the data and access pattern it handles best.
- **Rigid schema**: genomic and wearable data change shape; every new field needs a migration.
- **Scaling**: an RDBMS scales mostly vertically; 100 million patients outgrow one server.
- **Slow joins**: images, time series and relationships forced into tables cause 30 second queries.

Data	Store
Billing, prescriptions	Relational (ACID)
Patient profiles	Document (MongoDB)
Wearable time series	Column-family (Cassandra)
Recent vitals cache	Key-value (Redis)
Trials, referrals	Graph (Neo4j)
X-rays, MRIs	Object storage + metadata

- **Costs**: more systems to run and secure; keep stores in sync with Kafka or change data capture.
- **Compliance in every store**: encryption, access control, audit logs, EU data kept in EU regions.

Remember: right store per data type, sync, compliance everywhere.

(b) Four NoSQL types (10 marks)

Type	Model	Example	Healthcare use	Weak at
Key-value	key to opaque value	Redis, DynamoDB	latest vitals, sessions	queries not on the key
Document	nested JSON	MongoDB	patient profiles, imaging metadata	joins, multi-document transactions
Column-family	rows by key, column families	Cassandra, HBase	monitoring streams, genomic variants	ad hoc queries
Graph	nodes and edges	Neo4j	clinical trials, drug interactions	bulk scans

- **Profiles**: one read returns the whole record; new fields need no migration.
- **Imaging**: metadata in documents, files in object storage.

Remember: four models, one product each, one use each, one weakness each.

Question 3 (20 marks): Fraud rings with Spark GraphX

Key facts: millions of claims, about 10 percent fraudulent, about 1 billion dollars lost a year; rules miss rings that share addresses, phones and providers.

(a) GraphX concepts and the claims graph (10 marks)

- **Property graph**: directed multigraph with properties on vertices and edges.
- **VertexRDD**: `(VertexId, properties)` for people, claims, providers, addresses, phones.
- **EdgeRDD**: `Edge(src, dst, property)` such as FILED, LIVES_AT, REPAIRED_BY, USES.
- **Triplets**: an edge with both its vertices, e.g. "claimant X used shop Y".
- **Connected components** find large linked groups mixing many claimants with few addresses.
- **Triangle counting** finds tight clusters typical of staged accidents.
- **APIs**: GraphX is Scala and Java only; use GraphFrames from Python.

```
Person A --LIVES_AT--> Address 1 <--LIVES_AT-- Person B
   | FILED                                    | FILED
Claim 1 --REPAIRED_BY--> Shop X <--REPAIRED_BY-- Claim 2
```

- **Real time**: graph runs in batch every few hours; new claims are scored at once against its results.

Remember: vertices, edges, triplets, connected components.

(b) PageRank (10 marks)

- **Measures importance**: a vertex scores high when many, or important, vertices link to it.
- **Formula**: PR(v) = (1 - d)/N + d x sum of PR(u)/outdeg(u) over vertices u linking to v, with d = 0.85.
- **Iterative**: every vertex starts equal and shares its score over its links until changes fall below a tolerance (10 to 20 rounds).
- **Pregel**: each round is a superstep; in GraphX `graph.pageRank(0.0001)`.
- **High scores** point to ring organisers, corrupt clinics or repair shops, and shared phones or bank accounts.
- **Use**: run per suspicious component, send top scores to investigators, and use as a model feature.
- **Guard**: a busy legitimate hospital also scores high; compare with its normal volume.

Remember: d = 0.85, convergence, organisers and providers, prioritise investigations.

Question 4 (20 marks): Governance and data quality in healthcare

Key facts: 60 countries, 50 million patients, two breaches, over 75 million dollars in fines, 12 percent bad records, audits take 4 months.

(a) Pillars of data governance (8 marks)

- **Ownership**: a governance council, a data owner per domain, stewards who fix issues.
- **Policies and standards**: HL7 FHIR, ICD-10, LOINC; retention rules per country.
- **Data quality**: checks at ingestion and a quality score per source.
- **Security and privacy**: encryption, pseudonymised ids, masking, role-based access.
- **Compliance**: HIPAA minimum necessary; GDPR consent, erasure, 72-hour breach reporting; data residency.
- **Metadata and lineage**: a catalog tracing every number back to its hospital.
- **Audit**: immutable access logs cut audits from 4 months to days.
- **Balance**: analysts use de-identified data by default; identifiable access is approved and logged.

Remember: ownership, privacy, compliance, lineage, audit.

(b) Six data quality dimensions (12 marks)

Dimension	Meaning	Healthcare check
Accuracy	matches reality	lab values within clinical ranges
Completeness	required data present	patient id, birth date, diagnosis not null
Consistency	same fact everywhere	birth date equal in EHR, lab and billing; standard units
Timeliness	available when needed	lab results within 1 hour, alerts in seconds
Validity	follows rules and formats	ICD-10 code exists; discharge after admission
Uniqueness	one record per patient	master patient index merges duplicates

- **Validate at ingestion** and send bad records to a quarantine table with the reason.
- **Profile daily** and show a quality score per hospital.
- **Detect anomalies**: sudden drops in volume or jumps in values reveal faulty feeds.
- **Fix at the source** so the 12 percent error rate falls; tools: Great Expectations or Deequ.

Remember: six dimensions, a check for each, quarantine, master patient index.

Question 5 (20 marks): Cloud cost optimisation

Key facts: clients on AWS, Google Cloud and Azure often spend 50 percent or more over budget.

(a) Cost drivers (10 marks)

- **Storage**: priced per GB-month plus per request; copies, no deletion and small files add up.
- **Compute**: usually the biggest share; clusters sized for peak and left idle at night.
- **Inefficient queries**: `SELECT *` on unpartitioned tables and skewed jobs waste hours.
- **Streaming**: runs 24/7, so waste repeats every hour.
- **Data transfer**: egress and cross-region or cross-zone traffic is charged per GB.
- **Managed service premiums** for managed Kafka, Spark and warehouses.
- **No visibility**: without tags per team, nobody owns the bill.
- **Example**: a 20-node cluster at 1 dollar per node-hour running 24/7 costs about 14,400 dollars a month, even if busy 6 hours a day.

Remember: storage, compute, transfer, visibility.

(b) Storage optimisation (10 marks)

Tier	Access	Example	Data
Hot	daily	S3 Standard	last 30 days
Warm	monthly	S3 Standard-IA, GCS Nearline	1 to 6 months
Cold	rarely	Glacier Instant, GCS Coldline	audit history
Archive	almost never	Glacier Deep Archive	legal retention

- **Tiering trade-off**: colder tiers cost far less per GB but charge for and delay retrieval.
- **Compression**: CSV and JSON to Parquet or ORC with Snappy or ZSTD, often 5 to 10 times smaller and cheaper to scan.
- **Lifecycle rules**: move data by age automatically and delete staging data after 7 days.
- **Deduplication**: one curated copy shared by teams, incremental loads instead of full copies.
- **By data type**: compact small files into 128 MB to 1 GB files; video is already compressed, so tier it.
- **Example**: 500 TB of JSON becomes about 100 TB of Parquet; moving 80 percent to cold cuts about 11,500 dollars a month to about 1,000.

Remember: tiering, Parquet, lifecycle rules, deduplication.

Question 6 (20 marks): Feature engineering with Spark MLlib

Key facts: rules miss about 15 percent of fraud, churn found too late, credit scoring takes over 48 hours, sub-second scoring wanted.

(a) The MLlib feature pipeline (10 marks)

- **StringIndexer** (estimator): maps each category to a number, e.g. merchant category.
- **OneHotEncoder**: turns indexes into 0/1 vectors so categories are not treated as ordered.
- **VectorAssembler** (transformer): joins all columns into one `features` vector.
- **StandardScaler**: rescales features so amounts up to 1,000,000 do not drown counts up to 50.
- **Missing data**: `Imputer` with the median, plus a "missing" flag column.
- **New categories**: `handleInvalid="keep"` stops unseen values from crashing scoring.
- **Fit on training data only** to avoid leakage; the same pipeline then scores new data.

```
indexer = StringIndexer(inputCols=cats, outputCols=[c + "_idx" for c in cats],
        handleInvalid="keep")
encoder = OneHotEncoder(inputCols=[c + "_idx" for c in cats], outputCols=[c + "_vec" for c in
        cats])
assembler = VectorAssembler(inputCols=nums + [c + "_vec" for c in cats], outputCol="raw")
scaler = StandardScaler(inputCol="raw", outputCol="features")
lr = LogisticRegression(labelCol="is_fraud", weightCol="class_weight")
model = Pipeline(stages=[imputer, indexer, encoder, assembler, scaler, lr]).fit(train_df)
```

Remember: estimator vs transformer, stage order, handleInvalid, no leakage.

(b) Why feature engineering matters (10 marks)

- **Better features often beat a better algorithm**: a model only learns what its features show.
- **Velocity**: transactions in the last hour catch rapid fraud.
- **Ratio to normal**: amount divided by the customer's average flags unusual spending.
- **Location and device**: distance from usual place and new device flag catch account takeover.
- **Churn signals**: fewer logins and falling balances flag customers before they leave.
- **Imbalance**: use class weights; report precision, recall, F1 and PR-AUC, not accuracy.
- **Drift and leakage**: retrain daily; never use fields known only after fraud is confirmed.
- **Serving**: train in Spark daily, serve through a low-latency model server for sub-second scores.

Remember: derived features, imbalance metrics, drift, serving layer.

MSCIT 9128 Final Exam, October 2025: Model Answers

Format: Question 1 compulsory (40 marks), choose 3 from Section B (20 marks each), 3 hours.

Question 1 (40 marks): Big Data precision farming

> Note: parts 2, 4, 5 and 7 say "city" although the scenario is farming; swap sensors for traffic cameras and the same answers apply.

1.1 Identify the 5 V's of Big Data in this project (5 marks)

- **Volume:** thousands of soil sensors plus drone and satellite images, terabytes per season.
- **Velocity:** sensor readings every few seconds; pest alerts needed the same day.
- **Variety:** sensor values, images, weather feeds, farm records.
- **Veracity:** faulty sensors, cloud cover on satellite images.
- **Value:** better yield forecasts, less water and fertilizer wasted, early outbreak warnings.

1.2 Describe how data sources, storage, processing and analysis fit into one architecture (6 marks)

- **Sources:** soil sensors, weather stations, drones, satellites, farm records.
- **Ingestion:** Kafka for sensor streams; bulk upload for images.
- **Storage:** HDFS or S3 data lake, Parquet by date and farm.
- **Processing:** Spark Streaming for live alerts; Spark batch for nightly cleaning and features.
- **Analysis:** Spark MLlib for yield and disease models; rules turn predictions into plans.
- **Serving:** dashboards for policymakers, SMS alerts for farmers.

1.3 Compare the suitability of Hadoop and Spark for this project (6 marks)

Point	Hadoop (HDFS + MapReduce)	Spark
Model	Batch only, writes to disk each step	In memory, batch and streaming
Speed	Slow for iterative work	Much faster for iterative jobs
Real-time	Not suitable	Structured Streaming with Kafka
Machine learning	No built-in library	MLlib
Storage	HDFS stores data cheaply	Stores nothing; reads HDFS or S3

Verdict: use both. HDFS stores, Spark processes.

1.4 Illustrate how Kafka could manage the real-time data streams (5 marks)

- **Producers:** sensor gateways, weather service, drone upload service.
- **Topics per data type:** `soil-moisture`, `weather`, `pest-traps`, `drone-image-events`.
- **Partitions by `farm_id`:** order kept per farm, farms processed in parallel.
- **Consumer groups:** alerting and archiving read the same topics independently.
- **Replication factor 3, 7-day retention:** no data loss on one broker failure; replay after a failed job.

```
Sensor gateways --> soil-moisture (by farm_id) --> Spark Streaming --> irrigation alert
Weather API     --> weather                    --> archiver --> HDFS / S3
```

1.5 Analyze the role of predictive and prescriptive analytics (6 marks)

- **Predictive (what will happen):** forecast yield per field from soil, weather and images.
- **Predictive:** warn of pest or disease outbreaks days ahead from humidity and trap counts.

- **Prescriptive (what to do):** irrigation timing and amount from predicted moisture and rain.
- **Prescriptive:** fertilizer dose per zone, less waste and runoff.
- **Prescriptive:** rank farms for inspection when an outbreak is likely.

1.6 Evaluate the risks to data veracity and recommend ways to improve data quality (6 marks)

- **Risks:** faulty sensors, missing readings, cloudy images, timestamp errors.
- **Validation at ingestion:** range checks (moisture 0 to 100%) and schema checks.
- **Anomaly detection:** z-scores or comparison with neighbouring sensors.
- **Imputation:** interpolate short gaps and mark them as filled.
- **Cloud masking:** combine several satellite passes.
- **Calibration:** regular sensor checks and an owner per source.

1.7 Propose one new application of spatial analytics in this project (6 marks)

Variable-rate management zones:

- **Combine** GPS soil readings, drone NDVI and yield history on a GIS map.
- **Cluster** each field into zones with similar needs (k-means on location and soil).
- **Hotspot analysis** (Getis-Ord Gi*) finds pest clusters.
- **Act:** send zone plans to GPS-guided tractors and drip systems.
- **Benefit:** inputs only where needed, outbreaks contained early.
- **Policy:** regional risk maps guide subsidies.

Remember: 5 V's, HDFS stores Spark processes, topics partitions consumer groups.

Question 2 (20 marks): Fraud detection at a local bank

2.1 Explain why stream processing might be preferred over batch processing (4 marks)

- **Timing:** fraud must be stopped before money leaves; nightly batch finds it too late.
- **Per transaction:** each one is scored within milliseconds to seconds.
- **Action:** block the card or ask for an OTP while the customer is at the till.
- **Patterns:** bursts of small test payments appear in short windows.

2.2 Classify the task as classification or regression and justify (4 marks)

- **Binary classification:** output is fraud or not fraud.
- **Supervised:** labels come from confirmed fraud cases and chargebacks.
- **Probability plus threshold:** a score such as 0.87 is compared with a cut-off.
- **Not regression:** no continuous number is predicted.

2.3 Identify two evaluation metrics suitable for imbalanced data (4 marks)

- **Accuracy fails:** with 0.1% fraud, "never fraud" scores 99.9% and catches nothing.
- **Recall** = TP / (TP + FN): share of real frauds caught.
- **Precision** = TP / (TP + FP): share of alerts that are real fraud.
- **Also:** F1, PR-AUC.

2.4 Discuss how real-time alerts could reduce losses (4 marks)

- **Block or hold** the payment before it completes.
- **Freeze the card** after the first suspicious use.

- **Confirm by SMS** so stolen cards are reported in minutes.
- **Lower costs:** fewer chargebacks, more trust.

2.5 Suggest one strategy to reduce false alarms (4 marks)

Tune the decision threshold using business costs:

- **Pick the threshold** where the cost of false alarms plus missed fraud is lowest, not a fixed 0.5.
- **Two levels:** high score blocks, medium score asks the customer to confirm.
- **Feedback loop:** analyst decisions become new labels; retrain regularly.

Remember: binary classification, recall and precision, threshold tuning.

Question 3 (20 marks): Predicting student dropout on an online learning platform

3.1 Describe how predictive analytics could identify at-risk students (4 marks)

- **Features per week:** logins, video completion, quiz scores, late assignments, days since last login.
- **Train** a classifier (logistic regression, random forest) on past dropped and completed students.
- **Score weekly:** high dropout probability flags the student.
- **Explain the flag** (e.g. no login for 10 days) so tutors can help.

3.2 Compare descriptive and predictive analytics in this context (4 marks)

	Descriptive	Predictive
Question	What happened?	What is likely to happen?
Example	18% dropped out last term	This student has a 75% dropout risk
Methods	Counts, averages, dashboards	ML models on past outcomes

3.3 Suggest one data visualization tool for tracking learning progress (4 marks)

Power BI (or Tableau):

- **Line charts** of weekly logins and quiz scores.
- **Heat map** of video completion shows where students give up.
- **Filters and drill-down** from class to student, with alerts when a metric drops.

3.4 Explain how ethical considerations apply when analyzing student data (4 marks)

- **Privacy and consent:** collect only what is needed, with consent.
- **Security:** limited access, encryption, access logs.
- **Fair use:** a risk score triggers support, never punishment.
- **Bias:** weak internet lowers logins, not commitment.

3.5 Propose ways to balance model accuracy and fairness (4 marks)

- **Remove sensitive attributes** and check for proxies such as device type.
- **Measure per group:** equal opportunity (recall, false positive rate) and demographic parity.
- **Fix gaps** by reweighting data or per-group thresholds, accepting a small accuracy loss.
- **Human in the loop:** tutors review flags; audit every term.

Remember: logins and quiz features, what happened vs what will happen, support not punishment.

Question 4 (20 marks): Machine learning on X-ray images in a hospital

4.1 Explain how deep learning can assist in image recognition tasks (4 marks)

- **CNNs** learn features directly from pixels.
- **Layers:** early ones detect edges, deeper ones detect nodules, opacities, fractures.
- **Transfer learning:** fine-tune a pre-trained network (ResNet, DenseNet) on fewer labelled X-rays.
- **Output:** probability per condition, optionally with age and symptoms.

4.2 Describe how data preprocessing improves image quality before modeling (4 marks)

- **Resize and crop** to a fixed size (e.g. 224 x 224); remove borders and text.
- **Normalize** pixel values so images from different machines are comparable.
- **Denoise and enhance contrast** (Gaussian filter, histogram equalization) to show faint lesions.
- **Augment** with small rotations, flips and brightness changes to reduce overfitting.

4.3 Identify challenges of model interpretability in healthcare (4 marks)

- **Black box:** doctors cannot see why the model says "disease".
- **Accountability:** doctors and regulators need checkable reasons.
- **Wrong signals:** the model may use a scanner artefact or hospital marker.
- **Tools are approximate:** Grad-CAM and SHAP need clinical checking.

4.4 Discuss how bias might appear in medical datasets (4 marks)

- **Sampling bias:** one hospital, scanner or population.
- **Under-representation:** few children, older patients or rare conditions.
- **Label bias:** radiologists disagree, or access to tests is unequal.
- **Historical bias:** groups diagnosed late in the past are labelled "healthy".

4.5 Suggest techniques to ensure fairness and transparency (4 marks)

- **Diverse data** from several hospitals and scanners.
- **Per-group metrics:** sensitivity and specificity by age, sex and site.
- **Explanations:** Grad-CAM heat map with each prediction; a model card stating limits.
- **Doctor decides:** log predictions for audit and monitor after deployment.

Remember: CNN and transfer learning, resize normalize augment, per-group metrics.

Question 5 (20 marks): Inventory and demand forecasting for a retail company

5.1 Explain how temporal analytics could improve demand forecasting (4 marks)

- **Trend and seasonality:** growth over months; weekends, paydays, holidays.
- **Lag features:** last week's sales, same week last year, weather, promotions.
- **Models:** ARIMA, Prophet, or gradient boosting with date features.
- **Result:** fewer stockouts before holidays, less overstock after.

5.2 Describe how spatial analytics can help identify regional demand patterns (4 marks)

- **Map sales** by branch and region on a GIS map.
- **Find hotspots**, e.g. cold drinks in warmer regions.
- **Link to local factors:** weather, population, income, competitors.

- **Act:** stock branches differently, pick new sites.

5.3 Differentiate between operational and descriptive analytics in this case (4 marks)

	Descriptive	Operational
Purpose	Summarize the past	Run daily actions
Example	Rice sales rose 15% in the North	Stock below reorder point, reorder sent
Timing	Weekly, monthly	Near real time

5.4 Propose how real-time dashboards could support decision-making (4 marks)

- **One screen:** live sales, stock and forecast per branch.
- **Alerts:** low stock, sales spikes, weak promotions.
- **Same-day action:** move stock or change prices.

5.5 Evaluate the limitations of relying only on analytics for strategic decisions (4 marks)

- **Past-based:** models fail after sudden change (pandemic, new competitor).
- **Data quality:** wrong stock counts give confident wrong answers.
- **Missing context:** brand, suppliers, staff knowledge.
- **Correlation is not causation:** combine analytics with human judgement.

Remember: trend seasonality lags, GIS hotspots, analytics plus judgement.

Question 6 (20 marks): Sentiment analysis and social media data

> Note: the scenario is bank fraud but all five parts ask about social media sentiment, so the answers follow the parts as asked.

6.1 Explain how sentiment analysis extracts opinions from social media data (4 marks)

- **Collect** posts by API, keyword or hashtag (e.g. mentions of the bank).
- **Clean** text: remove links, handle emojis and slang, tokenize.
- **Classify** as positive, negative or neutral (lexicon scores or ML models such as BERT).
- **Aggregate** by topic, time and place.

6.2 Discuss the benefits of combining text analytics with visualization tools (4 marks)

- **Fast reading:** trend lines, word clouds, topic charts.
- **Spikes stand out**, e.g. negative sentiment on the day the app fails.
- **Maps and filters** show where and who.
- **Shared dashboards** for all teams.

6.3 Suggest how real-time analytics can enhance marketing responses (4 marks)

- **Catch complaint waves** within minutes, before they spread.
- **Measure campaigns live** and adjust the same day.
- **Route complaints** to customer care for quick replies.

6.4 Assess the challenges posed by fake accounts and misinformation (4 marks)

- **Bots** distort sentiment up or down.
- **False stories** spread faster than corrections and damage trust.
- **Sarcasm and fake reviews** lower model accuracy.

6.5 Recommend measures to improve data reliability (4 marks)

- **Bot detection:** account age, posting rate, follower ratio, duplicate text.
- **Credibility weighting** and removal of duplicates and spam.
- **Fact-check** viral claims against trusted sources.
- **Validate** on hand-labelled samples; add survey data.

Remember: collect clean classify aggregate, act faster, bot detection.